The *KNOWLEDGESTORE*: an Integrated Framework for Ontology Population

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Inria, September 6, 2013

KnowledgeStore: Motivations

- Ontology Population from text
  - Knowledge Base Population task
  - Automatic extension of Wikipedia infobox
- A flexible infrastructure to fill the gap between structured and unstructured data
  - Knowledge in the information extraction loop
- Larger variety of facts (i.e. include events)
- Exploit cross-document co-reference
- Ontology Population as an incremental process

Outline

- The KnowledgeStore data model
  - Resources
  - Mention
  - Entities
- The Trentino Media experience
  - EvaluationOngoing work under the NewsReader project
- Conclusion

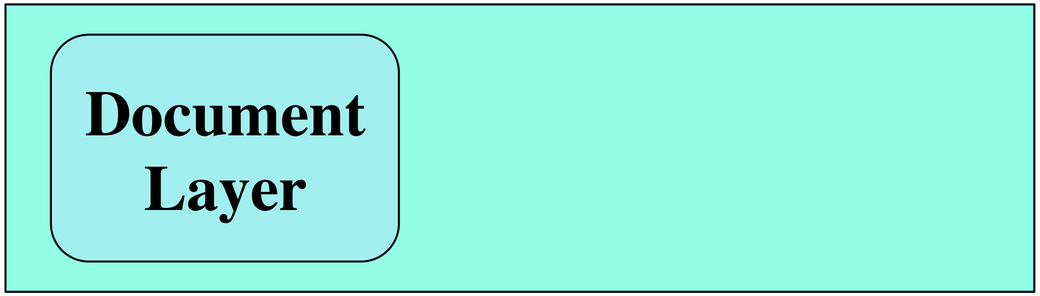
KnowledgeStore: Objectives

Design and implement an infrastructure, called **KnowledgeStore**

- Three-layer architecture: resource, mention, entity
- Allows to store massive amount of data about:
  - entities extracted from the data,
  - links to the data sources,
  - background domain knowledge
- Provides access via text-based search and semantic query languages
- Provides reasoning services over the knowledge it contains

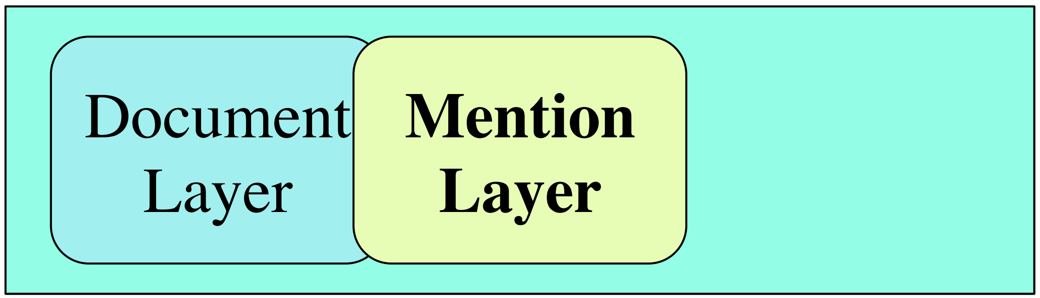
The Knowledge Store: content

- **Document Layer**
  - Textual resources (e.g. a newspaper article)
  - Linguistic Annotations of the textual resources (e.g. a document processed by a NLP pipeline)
  - Relevant Metadata (e.g. data source, language)



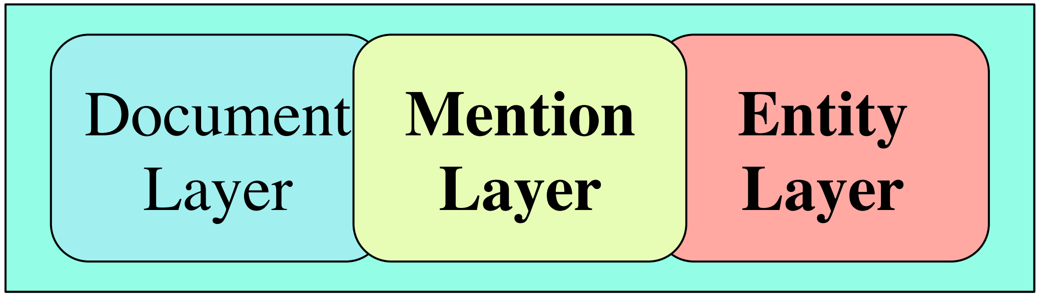
The Knowledge Store: content

- **Mention Layer**
  - Relevant portion of text (e.g. names of people and places)
  - Links to the data sources from which mentions have been extracted
  - Relevant Metadata (e.g. frequency of occurrence, confidence)



The Knowledge Store: content

- The **Entity Layer**
  - Unique Instances of mentions that have been extracted from the data
  - Coreference among mentions is resolved
  - Event reference schema
  - Links to Background domain knowledge
  - Relevant Metadata (e.g. factuality, provenance, confidence)



The Knowledge Store: content

Media: **R3**

title	**Rossi torna single** ***(Rossi is again single)***
creator	l'Adige
format	Text
date	2012/02/27
content	GOSSIP - **Il campione della Ducati Valentino Rossi** e la sua fidanzata Marwa Klebi si sono lasciati. … *(Tavullia champion Valentino Rossi and his girlfriend Marwa Klebi break up. …)*

Occurs in

Mention: **M2**

extent	Valentino Rossi
type	PER
firstname	Valentino
lastname	Rossi
start	33

References

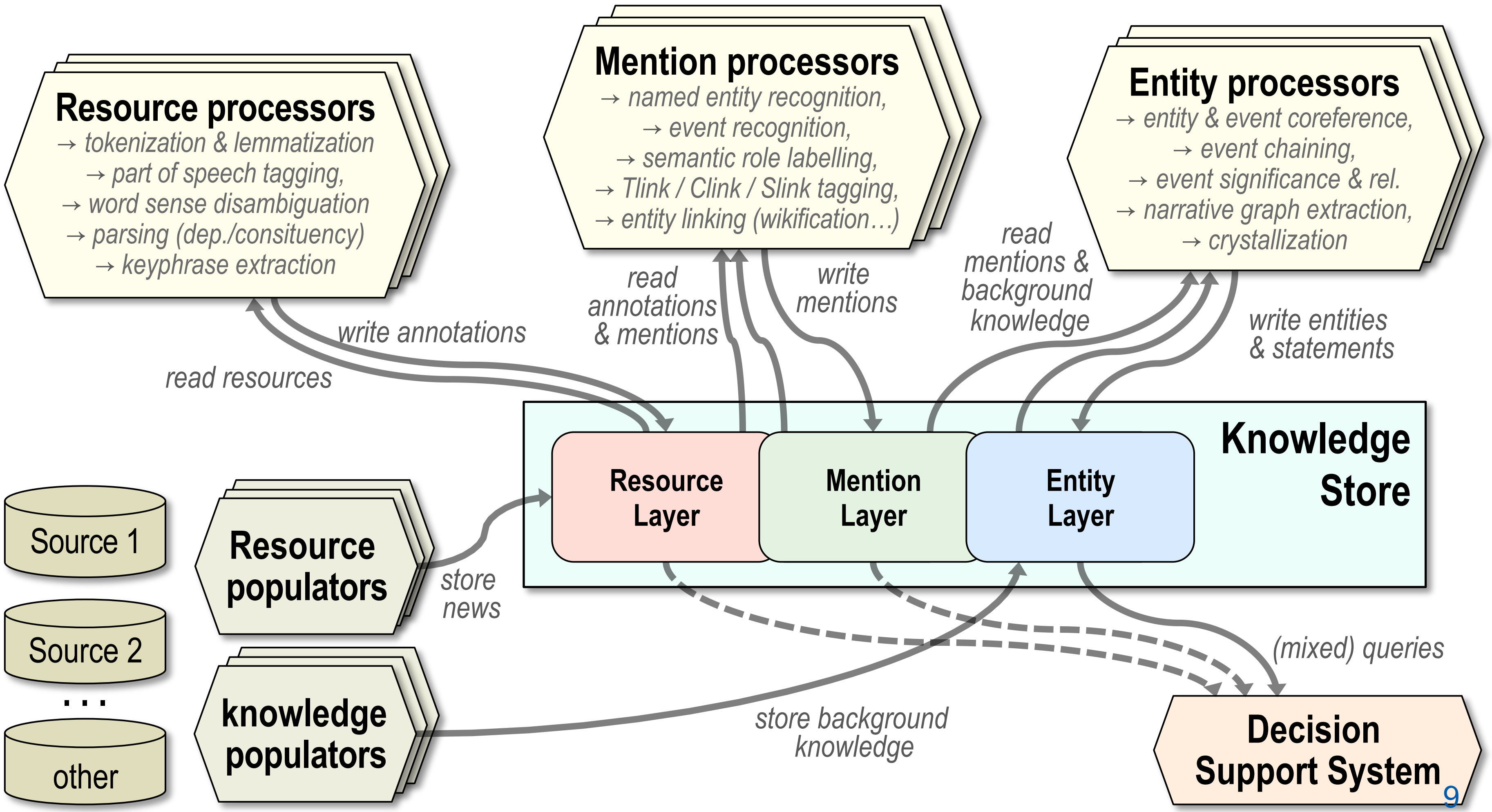
Entity: **E1**

Facts

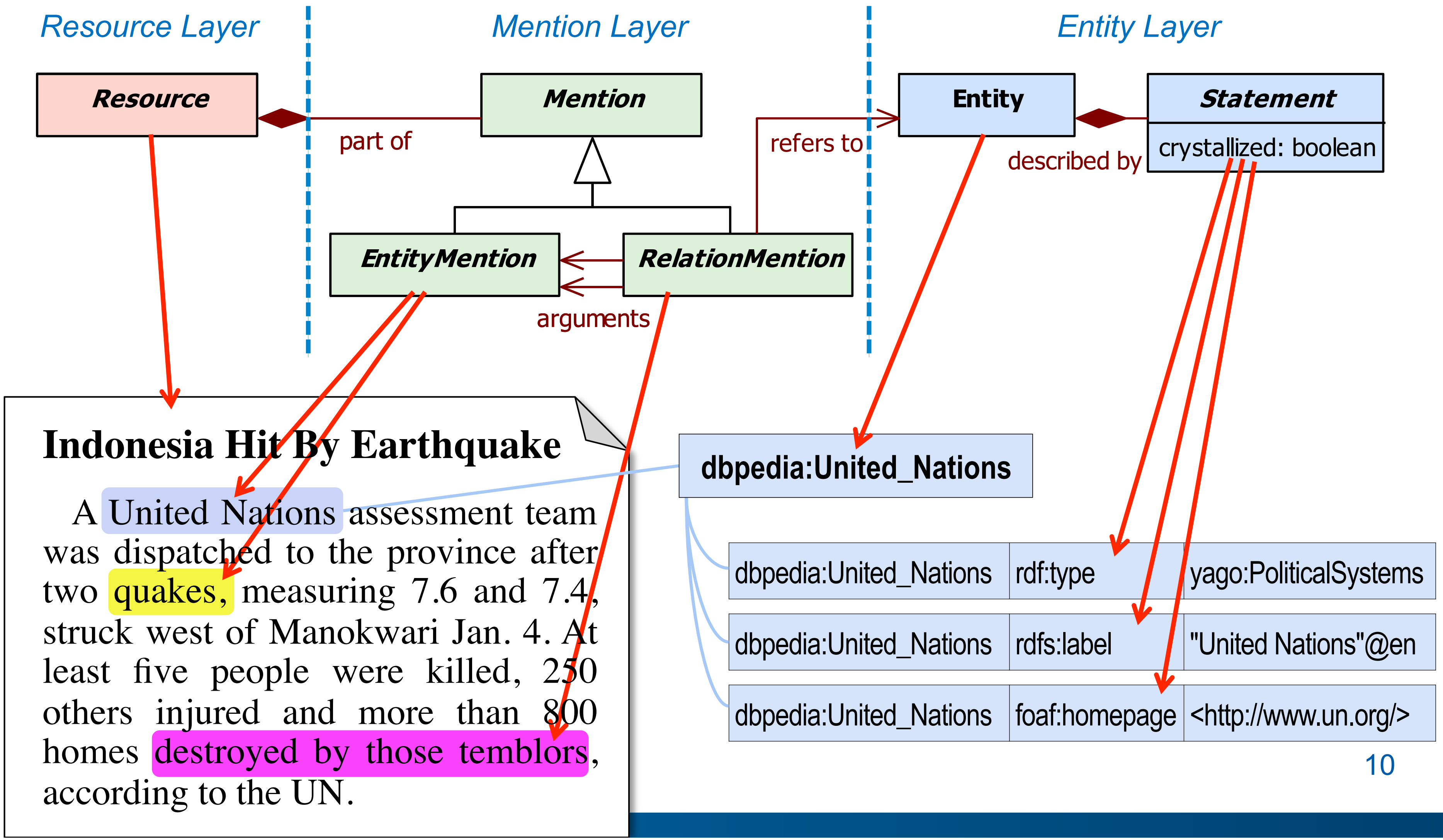
predicate	object	Source
type	Pilot	motogp.com
firstName	**Valentino**	**motogp.com, R3**
lastName	**Rossi**	**motogp.com, R3**
gender	**male**	**motogp.com, R3**
birthDate	1979-02-16	wikipedia.it
birthPlace	Urbino	wikipedia.it
height	182	motogp.com
weight	67	motogp.com
team	**Ducati**	**R3**

Document Layer | **Mention Layer** | Entity Layer

KnowledgeStore: General context



High Level Data Model



Resource annotation

news_00001.txt

Indonesia's West Papua Province Hit by Earthquake (Update2)

By Edward Johnson - January 7, 2009 01:55 EST

Jan. 7 (Bloomberg) -- Indonesia's West Papua province was hit by a magnitude 6.1 earthquake today, the latest powerful tremor to shake the region where five people were killed and hundreds injured at the weekend when buildings were destroyed.

The quake struck off the coast at 7:48 a.m. local time, 75 kilometers (50 miles) west of the region's main city of Manokwari, the U.S. Geological Survey said in an alert. There were no immediate reports of damage or casualties and no tsunami alert was issued.

A United Nations assessment team was dispatched to the province after two quakes, measuring 7.6 and 7.4, struck west of Manokwari Jan. 4. At least five people were killed, 250 others injured and more than 800 homes destroyed by those temblors, according to the UN.

About 14,000 people fled their homes at the weekend after a local tsunami warning was issued, the UN said on its Web site. The Indonesian government has transported relief supplies, including food, medicine, tarpaulins, generators and water purification equipment to the affected areas.

Two three-story hotels in Manokwari, a city of 160,000 people, collapsed in the Jan. 4 quakes and buildings were also damaged in the town of Sorong, according to the World Health Organization. Manokwari's Rendai Airport was also temporarily closed on security and safety concerns, the WHO said.

Indonesia lies in a zone where the Indo-Australian, Eurasian, Philippine and Pacific plates meet and occasionally shift, causing earthquakes and sometimes generating tsunamis. There have

News	nwr:news_00001
dc:title	Indonesia's West Papua Province Hit by Earthquake (Update2)
dc:creator	bloomberg.com
dc:language	EN
dc:issued	2009-01-07T01:55:00-05:00
nfo:wordCount	287
nfo:characterCount	1778
file	news_00001.txt
…	…

Resource linguistic annotation

ann_00001.txt

```
<token id="315" sentence="10" number="314">,</token>
<token id="316" sentence="10" number="315">India</token>
<token id="317" sentence="10" number="316">,</token>
<token id="318" sentence="10" number="317">Thailand</token>
<token id="319" sentence="10" number="318">and</token>
<token id="320" sentence="10" number="319">other</token>
<token id="321" sentence="10" number="320">countries</token>
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<Markables>
<C-SIGNAL id="1" comment="" >
<token_anchor id="135"/>
</C-SIGNAL>
<ENTITY_MENTION id="17" syntactic_type="NOM" head="people" comment="" reference_type="SPC" >
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</ENTITY_MENTION>
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<token_anchor id="131"/>
<token_anchor id="132"/>
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</ENTITY_MENTION>
<ENTITY_MENTION id="2" syntactic_type="NAM" head="UN" comment="" reference_type="SPC" >
<token_anchor id="141"/>
<token_anchor id="142"/>
</ENTITY_MENTION>
<ENTITY_MENTION id="3" syntactic_type="WHQ" head="where" comment="" reference_type="SPC" >
<token_anchor id="23"/>
</ENTITY_MENTION>
<ENTITY_MENTION id="4" syntactic_type="NAM" head="West Papua" comment="" reference_type="SPC" >
<token_anchor id="3"/>
```

Annotation	nwr:ann_00001
tool	nwr:textpro
annotationFormat	TAF
dc:issued	2009-01-08T05:55:00+01:00
file	ann_00001.txt
dc:source	nwr:news_00001
…	…

FONDAZIONE BRUNO KESSLER

Resource linguistic annotation

News	nwr:news_00001
dc:title	Indonesia's West Papua Province Hit by Earthquake (Update2)
dc:creator	bloomberg.com
dc:language	EN
dc:issued	2009-01-07T01:55:00-05:00
nfo:wordCount	287
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file	news_00001.txt
…	…

Annotation	nwr:ann_00001
tool	nwr:textpro
annotationFormat	TAF
dc:issued	2009-01-08T05:55:00+01:00
file	ann_00001.txt
dc:source	nwr:news_00001
…	…

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Entity mentions (organizations)

ORG	nwr:orgmen_00001
nif:beginIndex	146
nif:endIndex	160
nif:anchorOf	United Nations
foaf:name	United Nations
…	…

A United Nations assessment team was dispatched to the province after two quakes, measuring 7.6 and 7.4, struck west of Manokwari Jan. 4.

At least five people were killed, 250 others injured and more than 800 homes destroyed by those temblors, according to the UN

- Non-event entities
- Events
- Time expressions
- TLink signals
- CLink signals

ORG Mention	nwr:orgmen_00002
nif:anchorOf	the UN
head	UN
foaf:name	UN
…	…

Entity mentions (event)

Event Mention	nwr:evmen_00001
nif:anchorOf	quakes
pred	quake
polarity	POS
pos	NOUN
morphology	regular
…	….

team was dispatched to the province after two quakes, measuring 7.6 and 7.4, struck west of Manokwari Jan. 4.

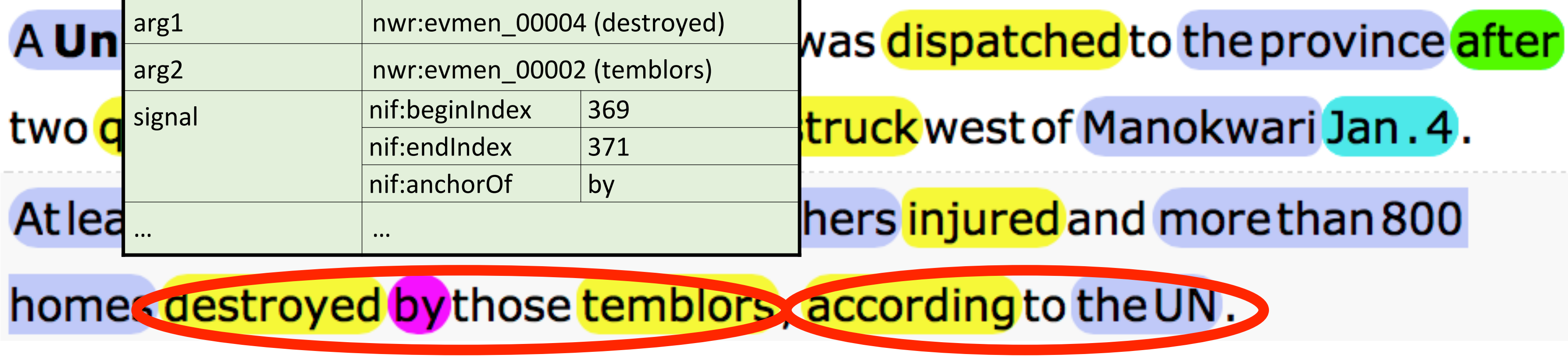
At least five people were killed, 250 others injured and more than 800 homes destroyed by those temblors,

Event Mention	nwr:evmen_00002
nif:anchorOf	temblors
pred	temblor
polarity	POS
pos	NOUN
morphology	regular
…	…

- Non-event entities
- Events
- Time expressions
- TLink signals
- CLink signals

Relation mentions (participation, c-link)

CLINK Mention	nwr:clmen_00001	
local	TRUE	
arg1	nwr:evmen_00004 (destroyed)	
arg2	nwr:evmen_00002 (temblors)	
signal	nif:beginIndex	369
	nif:endIndex	371
	nif:anchorOf	by
…	…	



Part. Mention	nwr:pmen_00001
local	TRUE
arg1	nwr:evmen_00007 (according)
arg2	nwr:orgmen_00002 (the UN)
semRole	UNDEF
…	…

- Non-event entities
- Events
- Time expressions
- TLink signals
- CLink signals

Mentions – linking to resource

News	nwr:news_00001
dc:title	Indonesia’s West Papua Province Hit by Earthquake (Update2)
dc:creator	bloomberg.com
dc:language	EN
dc:issued	2009-01-07T01:55:00-05:00
nfo:wordCount	287
nfo:characterCount	1778
file	news_00001.txt
….	….

ORG Mention	nwr:orgmen_00001	*“United Nations”*
dc:isPartOf	nwr:news_00001	

ORG Mention	nwr:orgmen_00002	*“the UN”*
dc:isPartOf	nwr:news_00001	

Event Mention	nwr:evmen_00001	*“quakes”*
dc:isPartOf	nwr:news_00001	

Event Mention	nwr:evmen_00002	*“temblors”*
dc:isPartOf	nwr:news_00001	

CLINK Mention	nwr:clmen_00001	*“destroyed by those temblors”*
dc:isPartOf	nwr:news_00001	

Part. Mention	nwr:pmen_00002	*“according to the UN”*
dc:isPartOf	nwr:news_00001	

Entities & statements

Entity	nwr:ent_00001

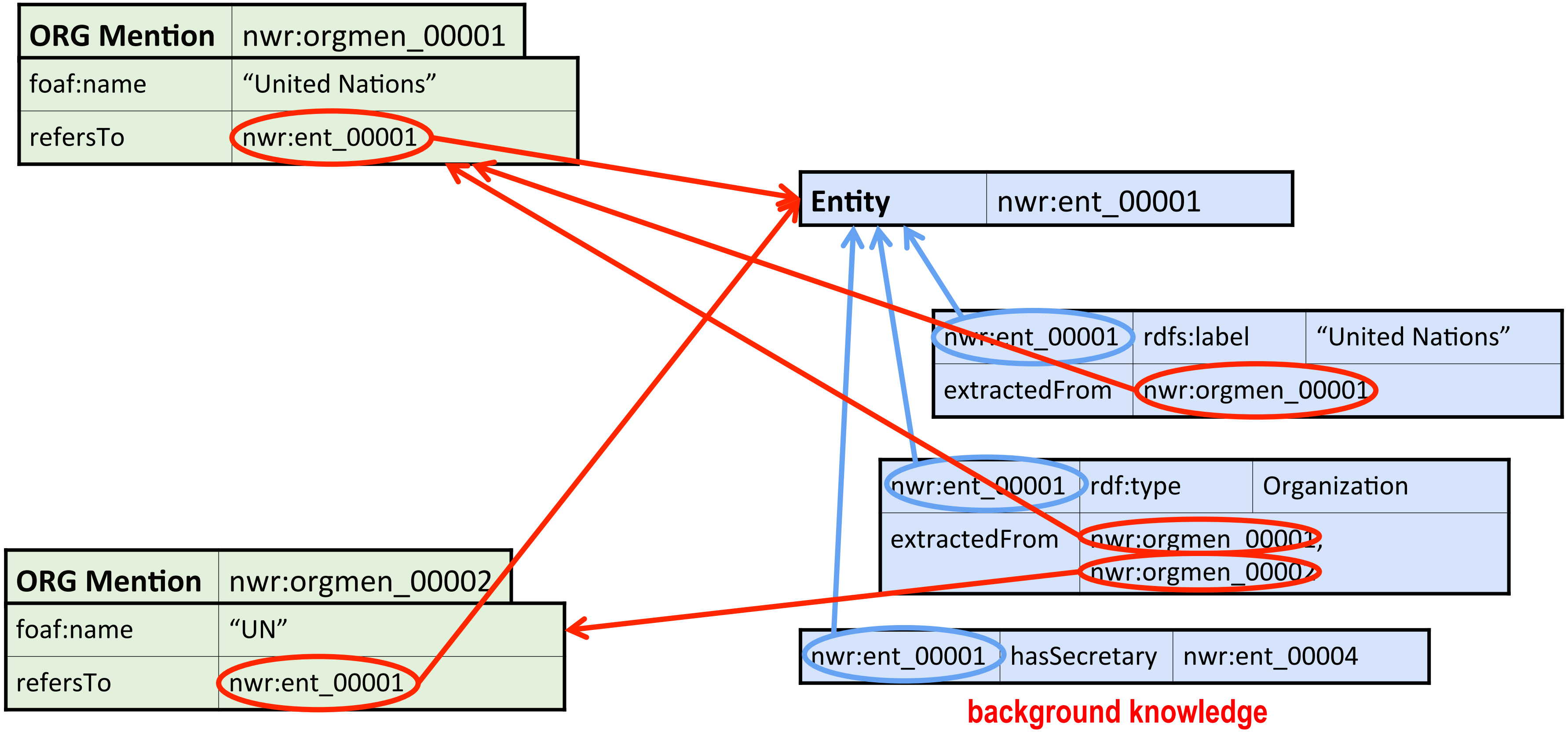
predicate	rdfs:label
subject	nwr:ent_00001
object	“United Nations”
context	nwr:context_00003
crystallized	false
confidence	0.4
…	…

predicate	owl:sameAs
subject	nwr:ent_00001
object	dbpedia:United_Nations
context	nwr:context_00003
crystallized	true
dc:source	dbpedia
…	…

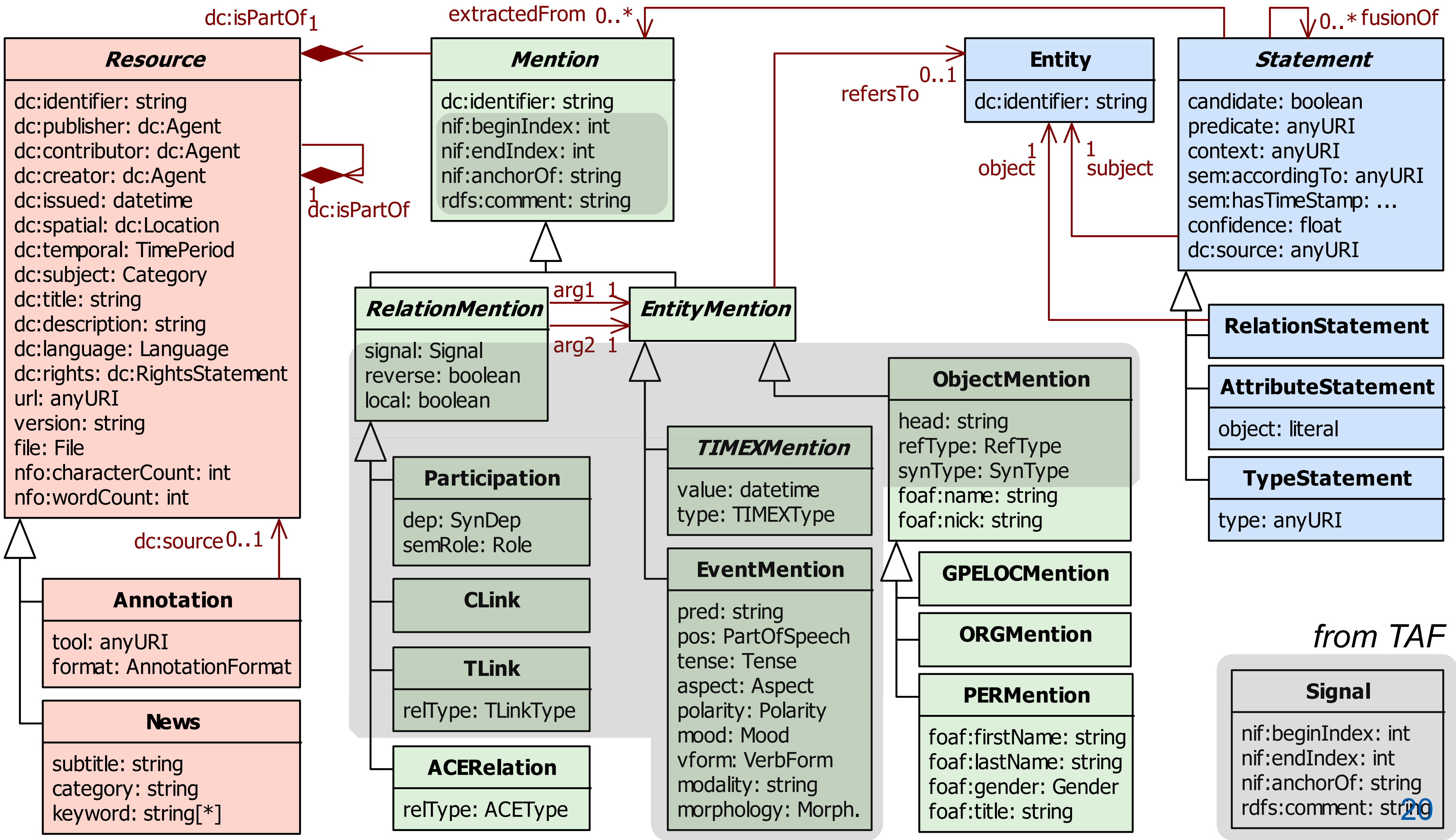
predicate	hasSecretary
subject	nwr:ent_00001
object	nwr:ent_00004
context	nwr:context_00003
crystallized	true
dc:source	nwr_BK
…	…

predicate	rdf:type
subject	nwr:ent_00001
type	Organization
context	nwr:context_00003
crystallized	true
confidence	1.0
…	…

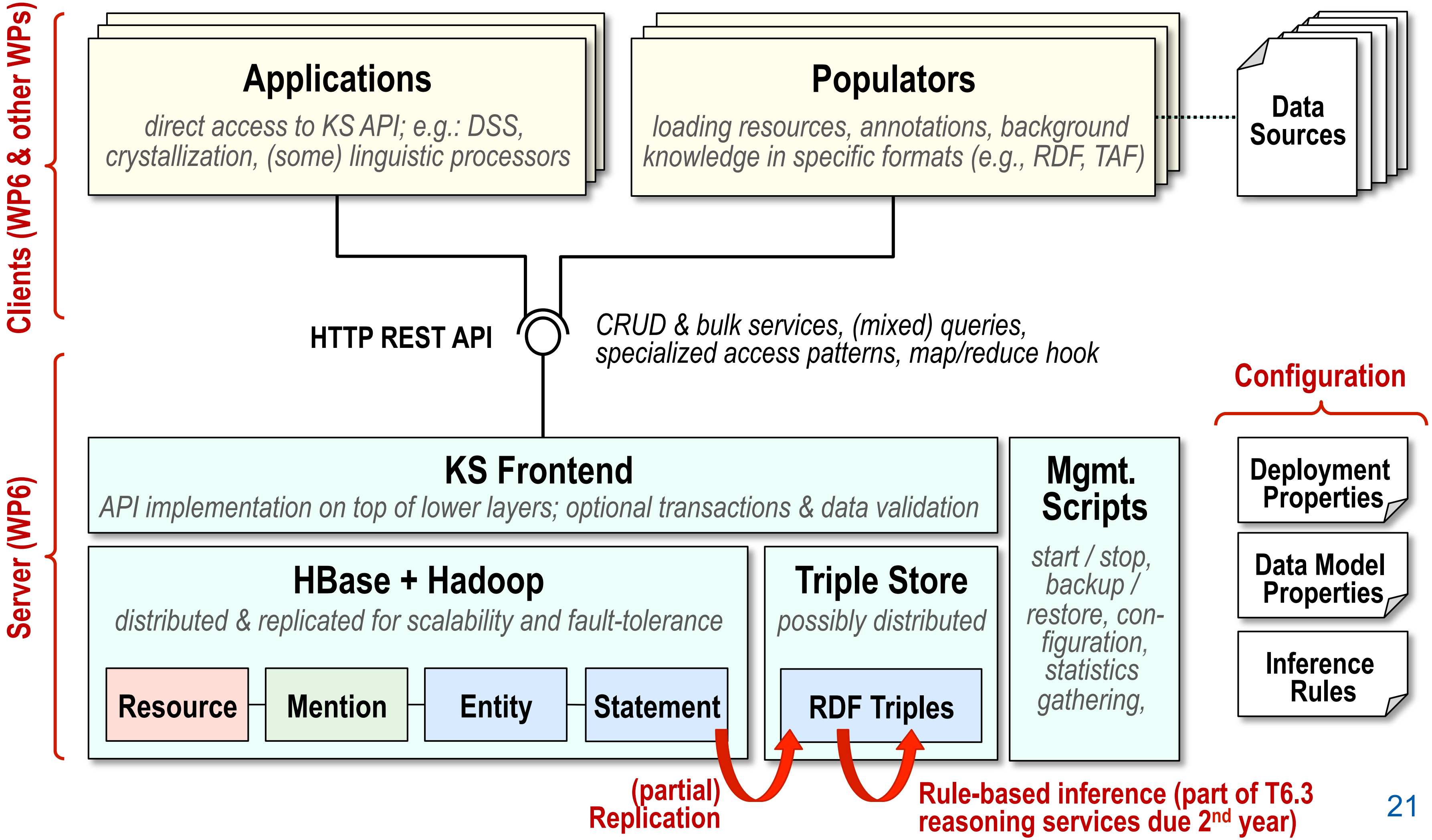
Entities & statements – linking to mentions



KnowledgeStore data model



High-level architecture



Trentino Media: Content Acquisition (1)

Content acquisition | Resource preprocessing | Mention extraction | Coreference resolution | Linking & entity creation

- Goal: acquire and load **news** and **background knowledge**
- Multimedia news
  - ~56 GB of textual news, images and videos in the Italian language
  - acquired from 4 news providers local to the Italian Trentino region
  - daily updated, since 1999

Provider	News	Images	Videos
l'Adige	733,738	21,525	-
VitaTrentina	33,403	14,198	-
RTTR	2,455	-	120 h
Fed. Cooperative	1,402	-	-
All	770,998	35,723	120 h

Trentino Media: Content Acquisition (2)

Content acquisition → Resource preprocessing → Mention extraction → Coreference resolution → Linking & entity creation

- Background knowledge
  - acquired from Italian Wikipedia, sport-related community sites, local and national level public administrations and government bodies
  - manual acquisition using ad-hoc Web site wrappers

Topic	Contexts	Persons	Organizations	Avg.properties	Facts (triples)
sport	136	8,570	191	3.81	192115
culture	20	9,785	1	2.00	33236
justice	7	354	10	2.16	1575
economy	7	49	1,203	4.47	11147
education	6	850	82	2.35	3573
politics	535	8,402	319	4.64	98780
religion	3	1,391	0	1.67	12855
total	714	28,687	1,806	3.64	352,244

Trentino Media: Resource Preprocessing

Content acquisition | **Resource preprocessing** | Mention extraction | Coreference resolution | Linking & entity creation

- Goal: **normalize and annotate news** so to ease further processing
- Several operations:
  - Conversion of multimedia resources to common formats
  - Segmentation of complex news
    - separation of individual stories in a news broadcast
    - separation of figures and captions in complex XML news articles
  - Automatic Speech Recognition
  - Annotations of news with linguistic taggers
    - part of speech tagging, based on TextPro tool [Pianta08]
    - temporal expression tagging, based on TextPro
    - key concept extraction, based on KX tool [Pianta10]

Trentino Media: Mention Extraction

Content acquisition → Resource preprocessing → **Mention extraction** → Coreference resolution → Linking & entity creation

- Mention extraction is based on the TextPro Mention Detection module
  - system based on the supervised training of a statistical model
  - training for the Italian language based on the Italian Content Annotation Bank (I-CAB) dataset [Magnini06]; measured F1 value: 82%
- Mention extraction statistics

News provider	PER mentions	ORG mentions	GPE/LOC mentions	Total mentions
l'Adige	5,387,994	3,100,994	3,052,011	11,540,999
VitaTrentina	144,486	100,789	136,611	381,886
RTTR	19,290	15,493	27,404	62,187
Fed. Coop.	14,404	12,731	8,513	35,648
All	5,566,174	3,230,007	3,224,539	12,020,720

Trentino Media: Coreference Resolution

Content acquisition | Resource preprocessing | Mention extraction | **Coreference resolution** | Linking & entity creation

- Goal: put mentions referring to the same entity in a **mention cluster**
  - **cross-document** (more complex) as mentions may belong to different news

… il presidente della Provincia di Trento Lorenzo Dellai ha lodato l'iniziativa per molteplici aspetti ...

Lorenzo Dellai
(PERSON entity)

... questa mattina il presidente Dellai ha incontrato le parti sociali sulla manovra che la Giunta sta mettendo a punto …
… e intanto l' Alto Adige è tra le regioni europee con il più basso tasso di disoccupazione

Entity type	PER	ORG	GPE/LOC	Total
Mention clusters	340,147	16,649	52,478	409,274

Trentino Media: Linking & Entity Creation

Content acquisition → Resource preprocessing → Mention extraction → Coreference resolution → **Linking & entity creation**

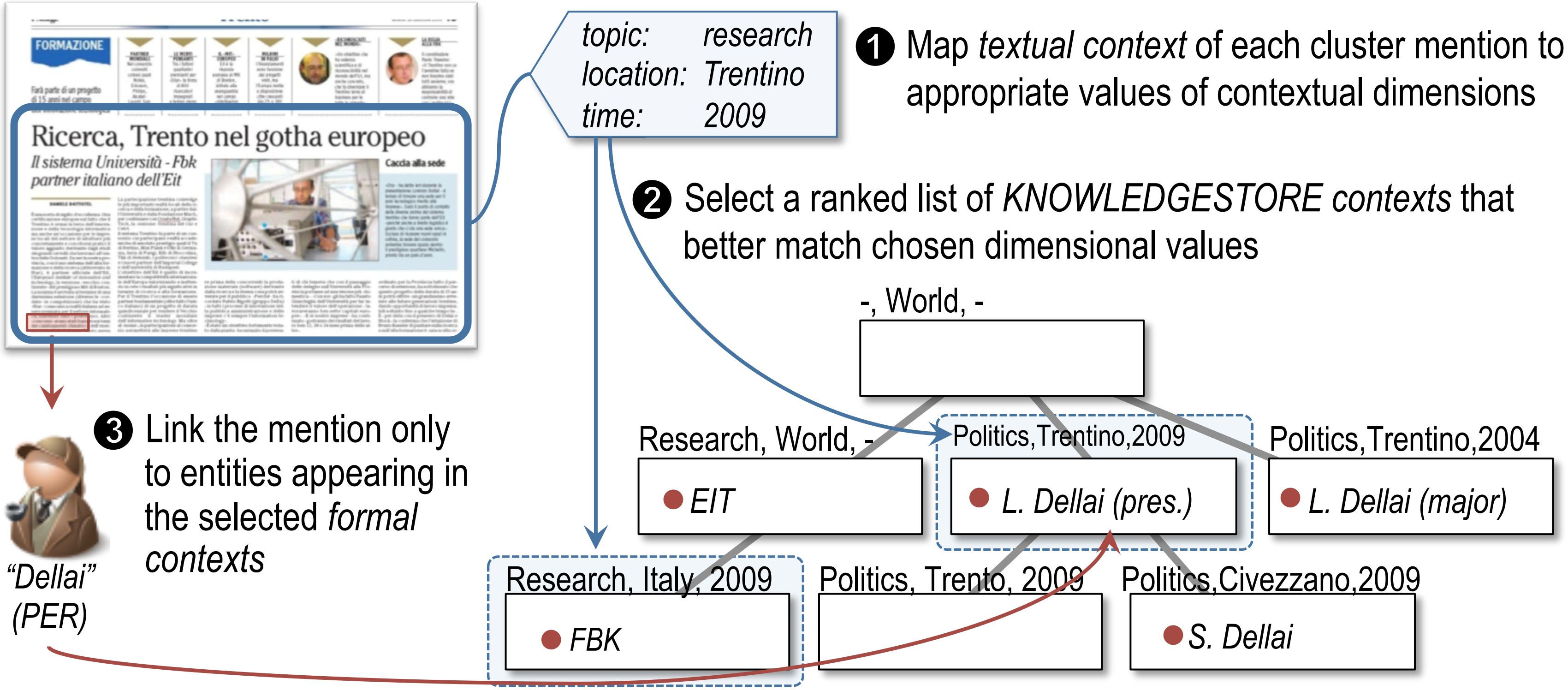
- Goal: link mention clusters to entities in the background knowledge and to external resources; create new entities from unlinked clusters
- Different types of linking are performed
  - **linking to GeoNames toponyms** of GPE/LOC clusters, using GeoCoder
  - **linking to background knowledge** entities of PER and ORG clusters
  - **linking to Wikipedia** pages of all the entities, using the WikiMachine tool
- New entities are created from unlinked clusters
- Linking and entity statistics:

Entity type	PER	ORG	GPE/LOC	Total
Linked clusters	5.03%	7.96%	48.64%	10.74%
Linked mentions	22.36%	12.02%	65.04%	31.03%
Resulting entities	321,713	17,129	52,478	421,320

Trentino Media: Linking & Entity Creation (2)

Content acquisition → Resource preprocessing → Mention extraction → Coreference resolution → **Linking & entity creation**

- Linking to background knowledge is context-driven [Tamilin10]
  - exploits the contextual organization of knowledge in the KNOWLEDGESTORE
  - 84,5% accuracy (i.e., correctly linked clusters) on gold standard of 298 clusters



Conclusion

- Our contributions
  - integration of techniques and technologies from different AI fields (NLP, Semantic Web, Machine Learning, …)
  - large scale application (770K news, 421K entities, 56 GB data)
  - tight interlinking of knowledge and multimedia
- Further research directions
  - extraction of (more complex) facts from news in the pipeline
  - expose extracted knowledge as Linked Data, to strengthen integration with Semantic Web
  - Application for the economical-financial domain in the EU NewsReader project

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